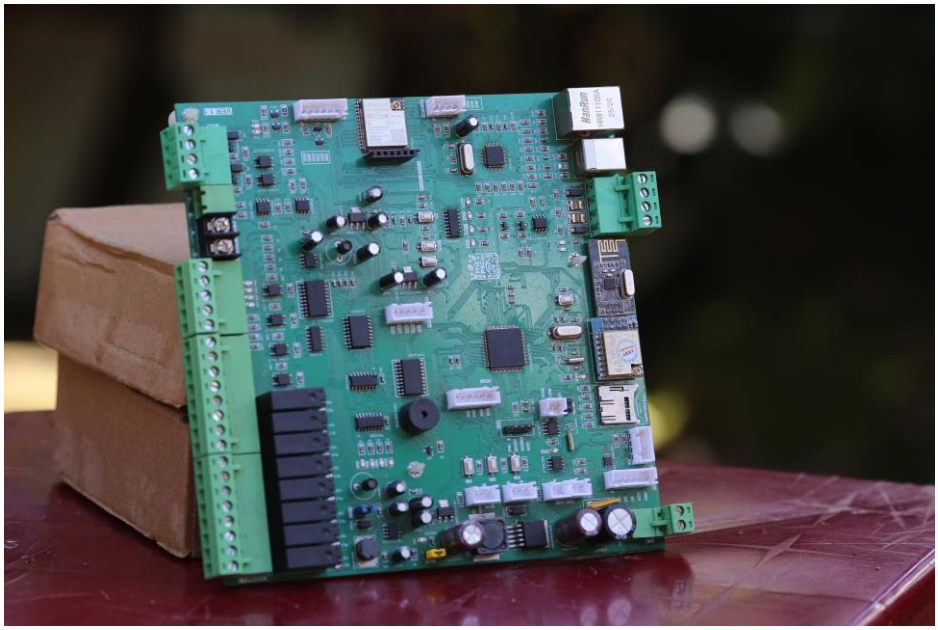
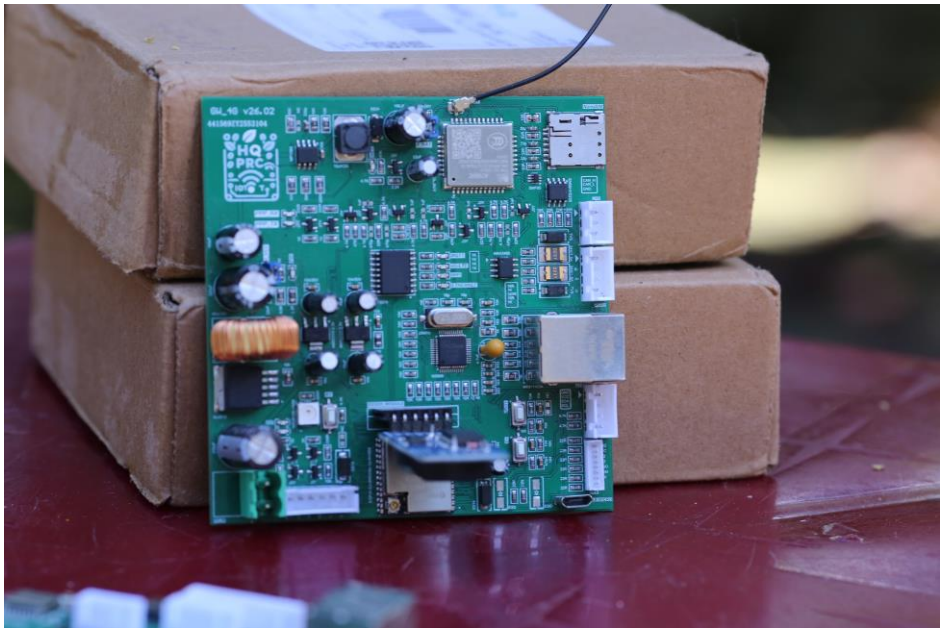
# HÌNH ẢNH 1 SỐ DỰ ÁN ĐÃ THỰC HIỆN

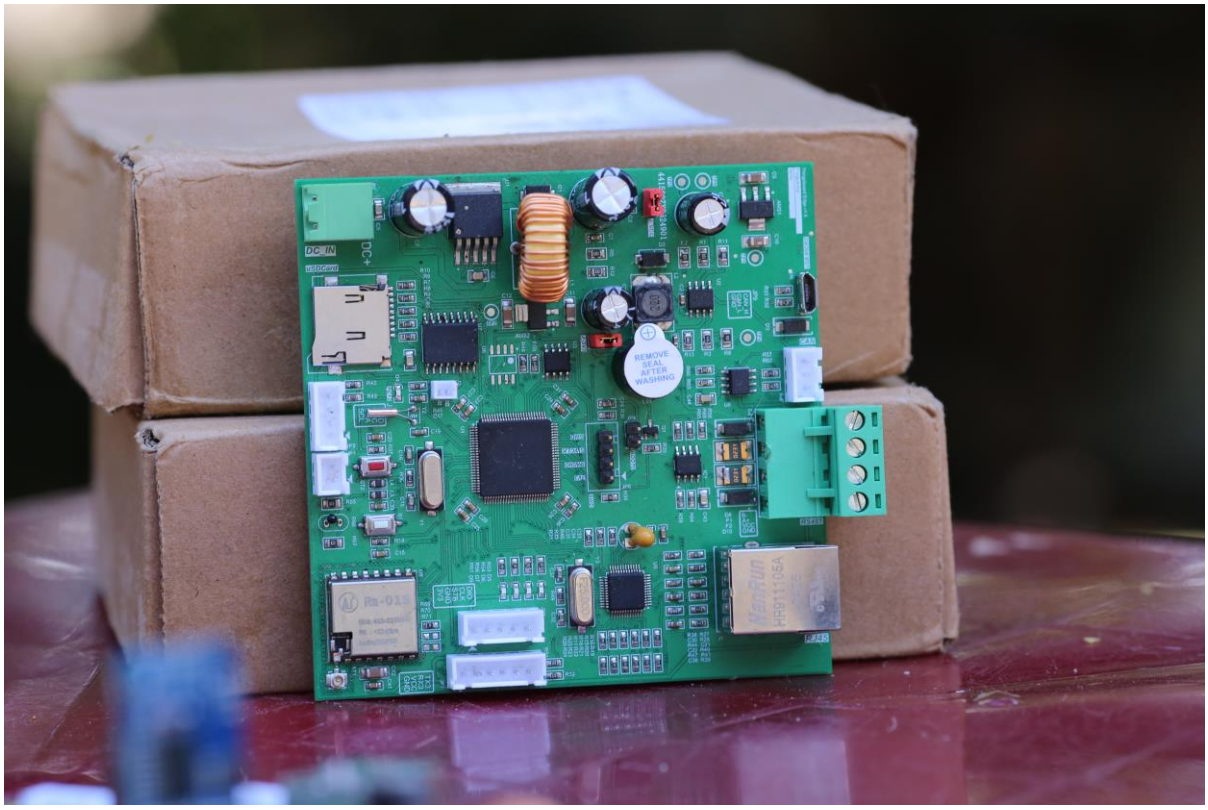
## 1. GATEWAY



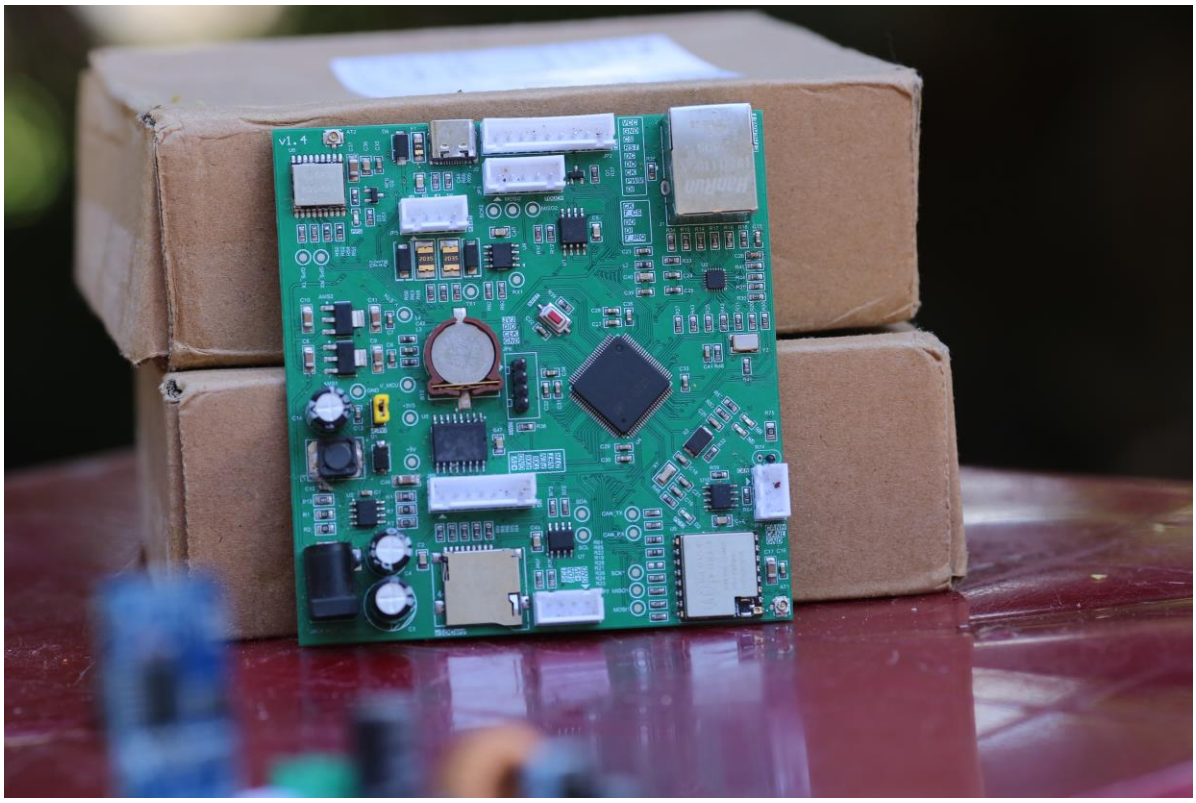
**Gateway ESP32+STM32F407(Uart)**



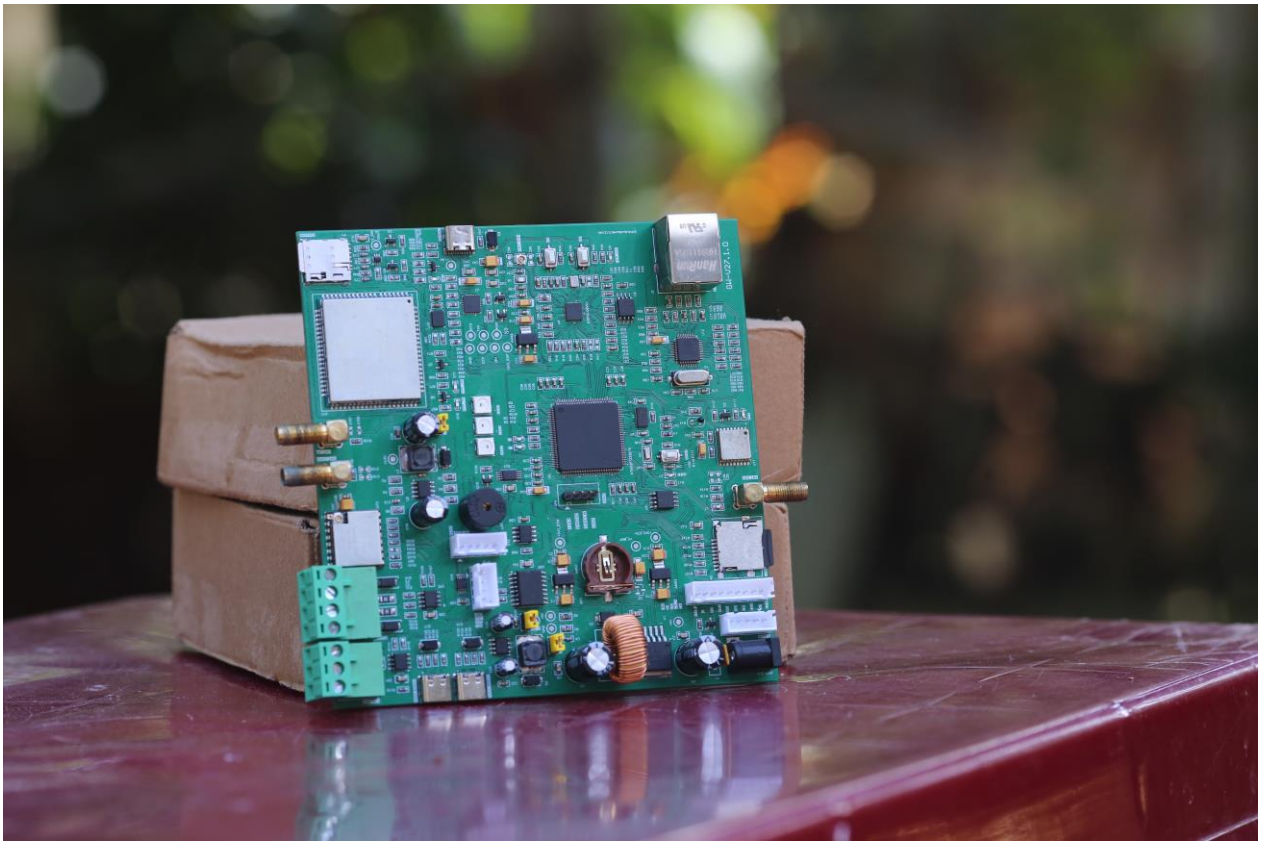
**Gateway ESP32S3 Ethernet/4G(SPI)**

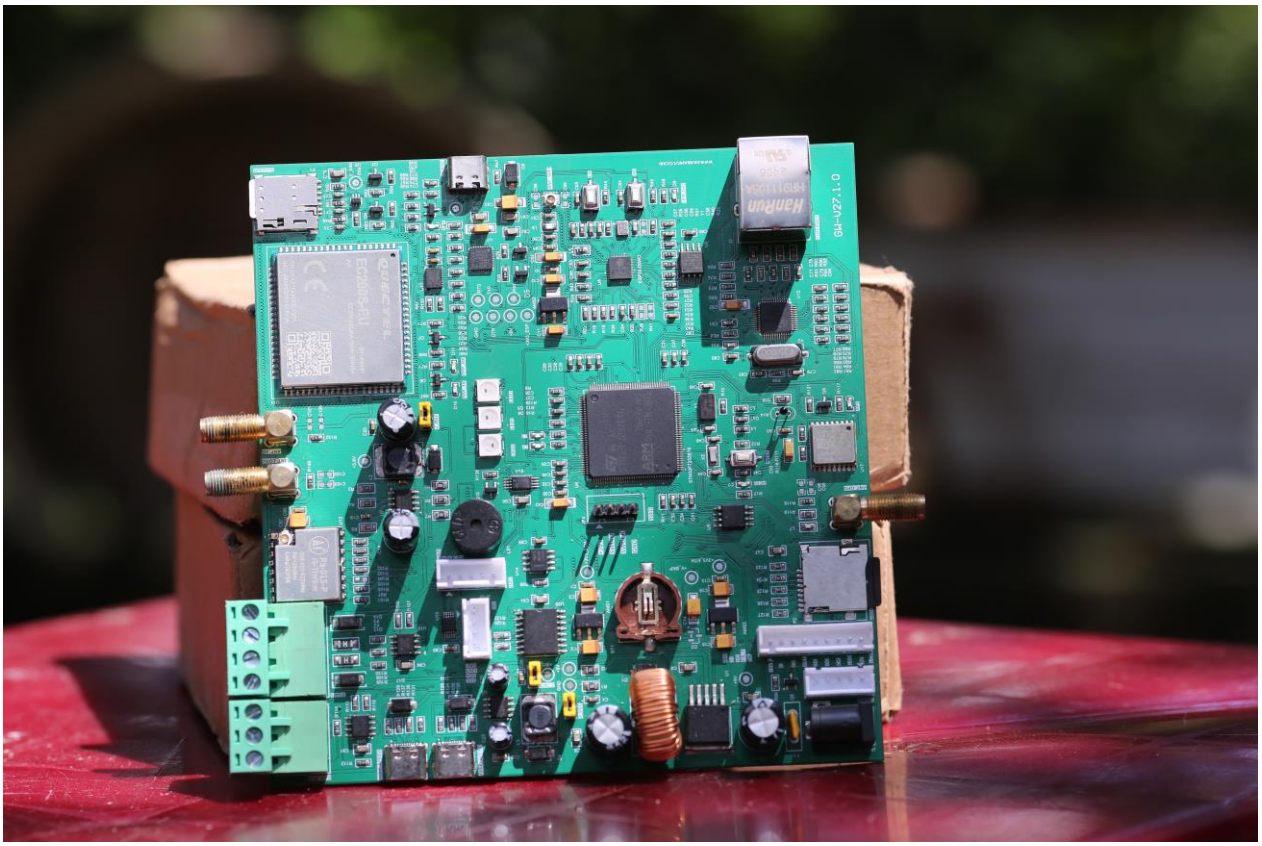



**Gateway STM32F407(Ethernet SPI)**



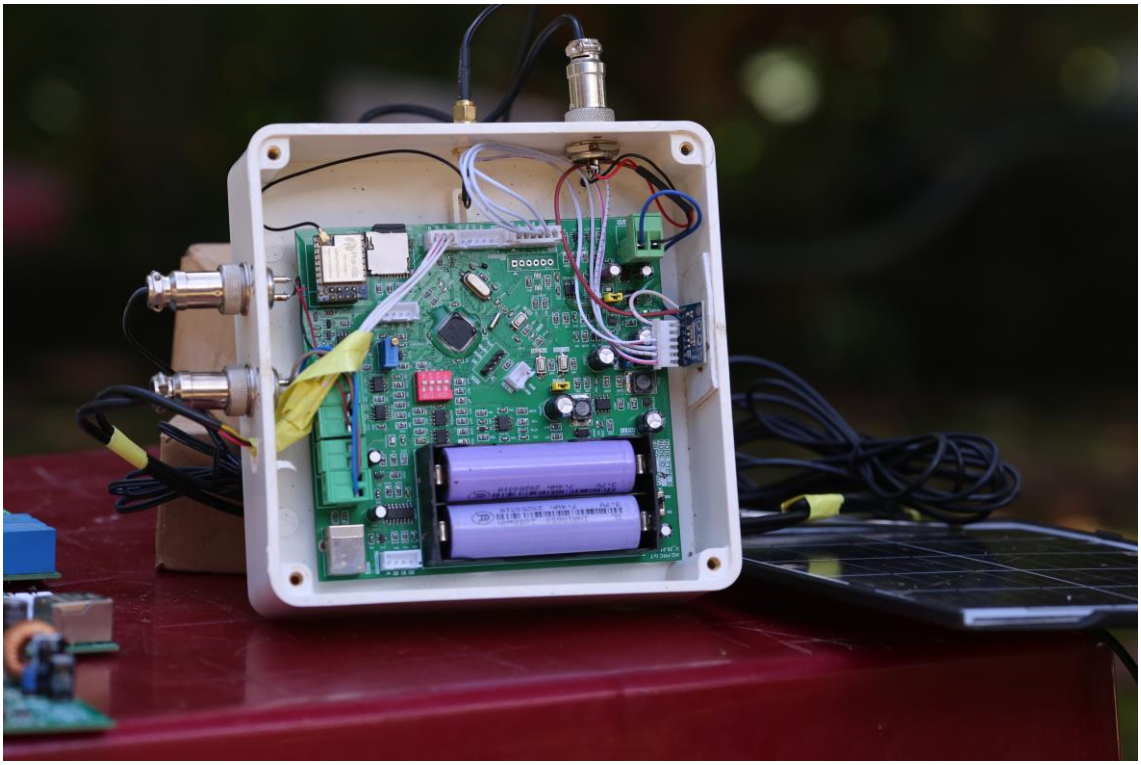

**Gateway STM32F407 (Ethernet RMII)**





**Gateway STM32F723+ESP32DOWD(SPI/UART)**

## 2. NODE CẢM BIẾN



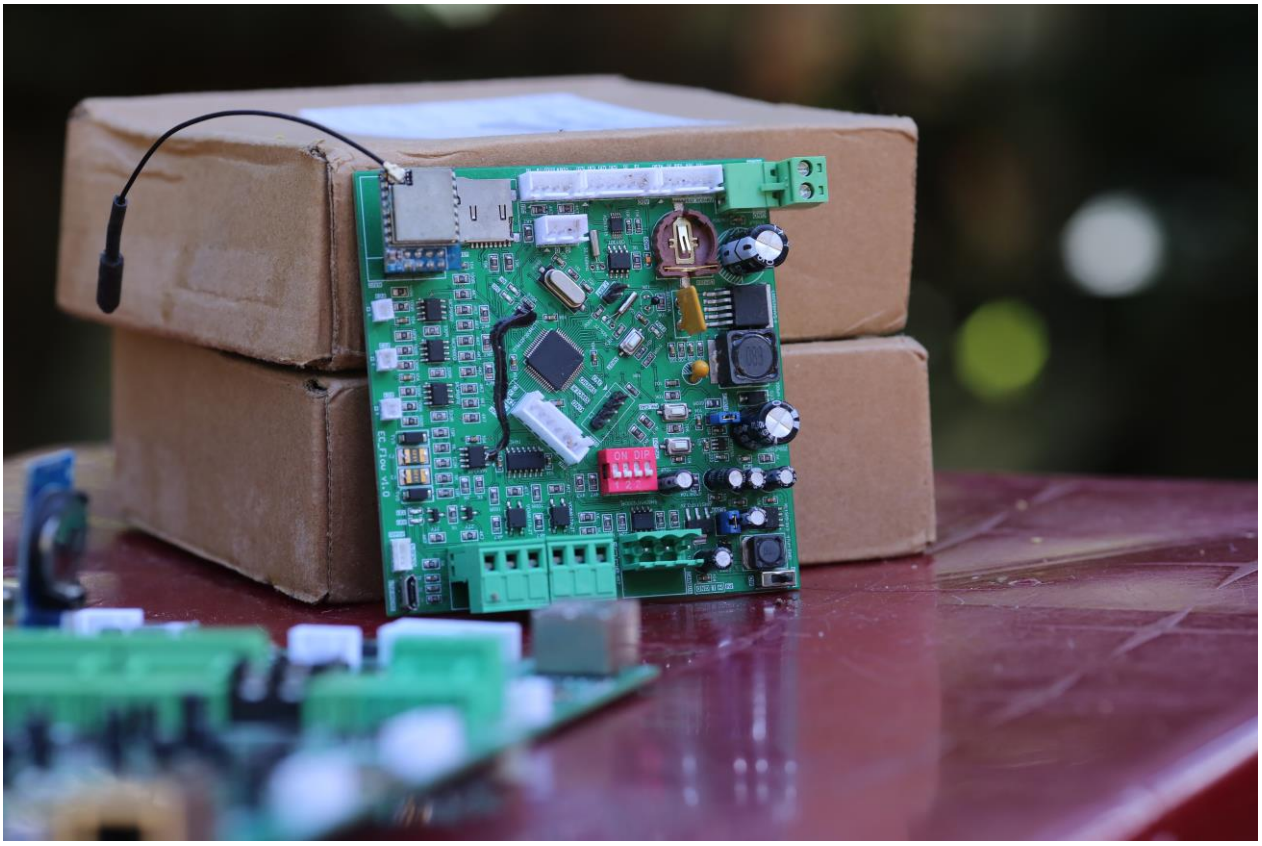
**Node LoRa EC/PH STM32F103RCT6**



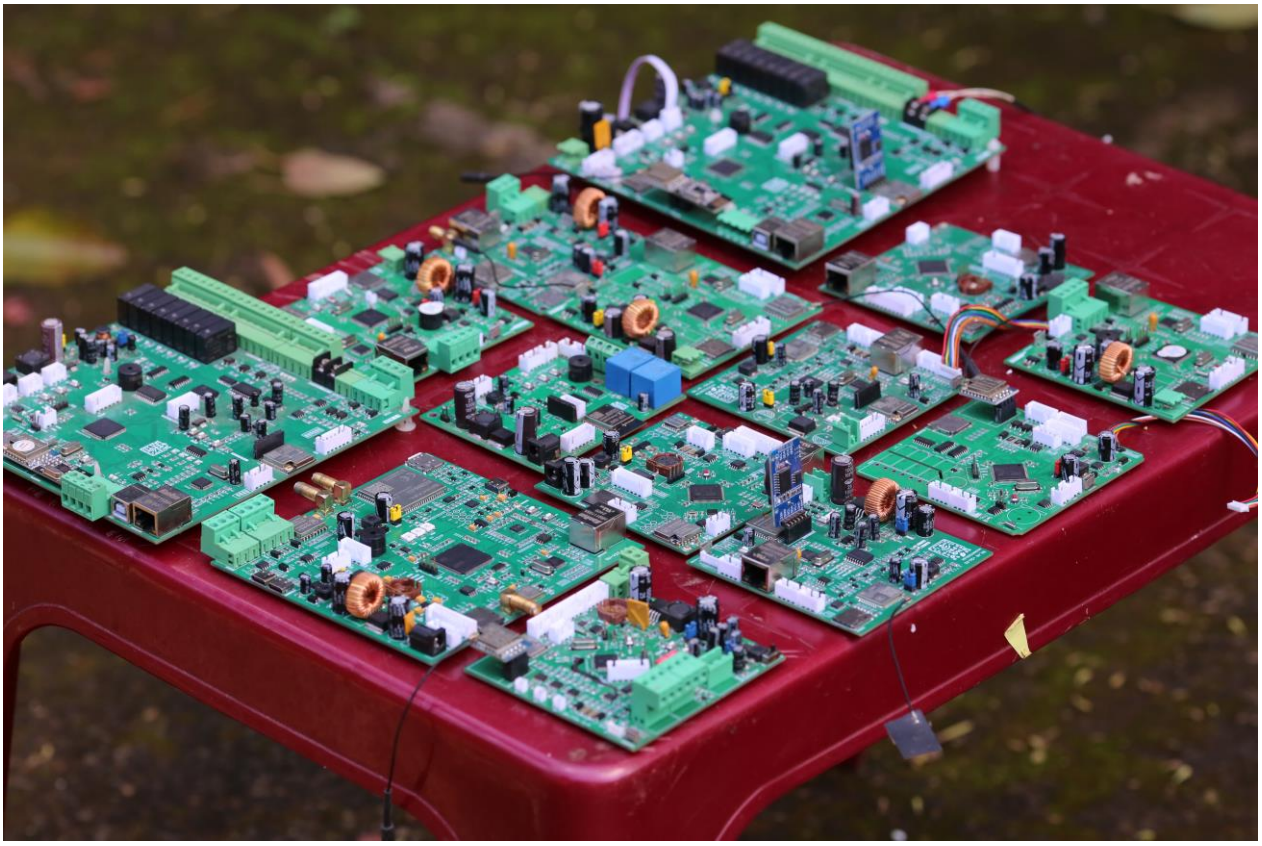

**Node LoRa khí hậu STM32F103C8T6**




**Node LoRa EC/RS485 STM32F411CEU6**



**Node EC Modbus RTU/CAN Bus**



**Các bo mạch đã thực hiện**